CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 1 of 10)

Based on the official CBSE blueprint, recent competency-based modifications, and the most heavily repeated questions from previous year papers (PYQs), here is Set 1 of the predicted Class 12 Maths Half-Yearly paper. This paper strictly adheres to the ~80% half-yearly syllabus (excluding Linear Programming and Probability).

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Write the principal value of $\cos^{-1}(\cos\frac{7\pi}{6})$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2023] [Confidence: Very High]

Q2. Find the principal value of $\sin^{-1}(\sin\frac{2\pi}{3})$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2022] [Confidence: Very High]

Q3. If A is a skew-symmetric matrix of order 3, what is the value of its determinant $|A|$? * **Chapter:** Matrices * **Source:** [PYQ 2020] [Confidence: Very High]

Q4. If A is a square matrix of order 3 and $|A| = 5$, find the exact value of $|adjA|$. * **Chapter:** Determinants * **Source:** [PYQ 2023] [Confidence: Very High]

Q5. What is the derivative of $2^{\sin x}$ with respect to x? * **Chapter:** Continuity and Differentiability * **Source:** [Practice] [Confidence: Medium]

Q6. The radius of a circle is increasing uniformly at the rate of 3 cm/s. What is the rate of change of the area of the circle with respect to time when the radius is 6 cm? * **Chapter:** Application of Derivatives * **Source:** [PYQ 2023] [Confidence: Very High]

Q7. Evaluate using properties of definite integrals: $\int_{-\pi/2}^{\pi/2} \sin^7 x\, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2019] [Confidence: Very High]

Q8. Evaluate the indefinite integral: $\int e^x(\sin x + \cos x)\, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2019] [Confidence: High]

Q9. Determine the order and degree (if defined) of the differential equation: $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$. * **Chapter:** Differential Equations * **Source:** [PYQ 2020] [Confidence: Very High]

Q10. Find the integrating factor of the linear differential equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$. * **Chapter:** Differential Equations * **Source:** [PYQ 2019] [Confidence: High]

Q11. Find the scalar projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$. * **Chapter:** Vector Algebra * **Source:** [PYQ 2023] [Confidence: Very High]

Q12. Write the direction cosines of the z-axis in 3D space. * **Chapter:** Three-Dimensional Geometry * **Source:** [Practice] [Confidence: High]

Q13. Given A is a square matrix of order 3 and $|3A| = k|A|$, find the value of k. * **Chapter:** Determinants * **Source:** [PYQ 2022] [Confidence: Very High]

Q14. Write the anti-derivative of the following function: $3x^2 + 4x^3$. * **Chapter:** Integrals * **Source:** [PYQ 2018] [Confidence: High]

Q15. The function $y = x^2 e^{-x}$ is strictly increasing in which interval? * **Chapter:** Application of Derivatives * **Source:** [PYQ 2020] [Confidence: Very High]

Q16. What is the value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{i} \times \hat{k}) + \hat{k} \cdot (\hat{i} \times \hat{j})$? * **Chapter:** Vector Algebra * **Source:** [PYQ 2016] [Confidence: Very High]

Q17. Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x^2$. Is this function one-one? * **Chapter:** Relations and Functions * **Source:** [PYQ 2019] [Confidence: High]

Q18. The direction ratios of a line are $2, -1, -2$. What are its direction cosines? * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2018] [Confidence: Very High]

Q19. (Assertion-Reason): Assertion (A): The function $f(x) = |x|$ is continuous everywhere but not differentiable at $x = 0$. **Reason (R):** Every continuous function is automatically a differentiable function. Choose the correct option. * **Chapter:** Continuity and Differentiability * **Source:** [Practice] [Confidence: High]

Q20. (Assertion-Reason): Assertion (A): The vectors $\vec{p} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{q} = 3\hat{i} + 2\hat{j} - 2\hat{k}$ are orthogonal. **Reason (R):** Two non-zero vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$. Choose the correct option. * **Chapter:** Vector Algebra * **Source:** [Practice] [Confidence: Very High]

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Evaluate: $\tan(\sin^{-1}(\frac{3}{5}) + \cot^{-1}(\frac{3}{2}))$. * **Chapter:** Inverse Trigonometric Functions * **Source:** [PYQ 2020] [Confidence: High]

Q22. Using determinants, find the value of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$, and $(k, 4)$ is 35 sq. units. * **Chapter:** Determinants * **Source:** [PYQ 2022] [Confidence: High]

Q23. Solve the differential equation: $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$. * **Chapter:** Differential Equations * **Source:** [PYQ 2019] [Confidence: Very High]

Q24. If $|\vec{a}| = 3$, $|\vec{b}| = 4$, and $|\vec{a} \times \vec{b}| = 6$, find the exact angle between the vectors $\vec{a}$ and $\vec{b}$. * **Chapter:** Vector Algebra * **Source:** [PYQ 2019] [Confidence: High]

Q25. Find the coordinates of the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+1}{4}$ intersects the xy-plane. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2018] [Confidence: High]

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R in the set of integers $\mathbb{Z}$ given by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation. * **Chapter:** Relations and Functions * **Source:** [PYQ 2019] [Confidence: Very High]

Q27. Express the matrix $A = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. * **Chapter:** Matrices * **Source:** [PYQ 2020] [Confidence: Very High]

Q28. Differentiate $y = x^x + (\sin x)^{\log x}$ with respect to x. * **Chapter:** Continuity and Differentiability * **Source:** [PYQ 2022] [Confidence: Very High]

Q29. Evaluate the indefinite integral involving a linear numerator and quadratic denominator: $\int \frac{x+2}{2x^2+6x+5}\, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2022] [Confidence: High]

Q30. Solve the linear differential equation: $(1+x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$. * **Chapter:** Differential Equations * **Source:** [PYQ 2019] [Confidence: Very High]

Q31. Find the shortest distance between the lines represented by $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. * **Chapter:** Three-Dimensional Geometry * **Source:** [PYQ 2019] [Confidence: Very High]

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Using the matrix method, solve the following system of linear equations: $2x + 3y + 3z = 5\; x - 2y + z = -4\; 3x - y - 2z = 3$ * **Chapter:** Matrices and Determinants * **Source:** [PYQ 2020] [Confidence: Very High]

Q33. Show that the semi-vertical angle of a right circular cone of maximum volume and of a given slant height is $\tan^{-1}\sqrt{2}$. * **Chapter:** Application of Derivatives * **Source:** [PYQ 2020] [Confidence: Very High]

Q34. Evaluate the following definite integral using properties of definite integrals: $\int_0^{\pi/2} \log(\sin x)\, dx$. * **Chapter:** Integrals * **Source:** [PYQ 2018] [Confidence: Very High]

Q35. Find the area of the region enclosed between the two parabolas $y^2 = 4x$ and $x^2 = 4y$. * **Chapter:** Application of Integrals * **Source:** [PYQ 2019] [Confidence: Very High]

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (School Awards) Three schools A, B, and C decided to organize a fair for collecting funds for awarding their students for the core values of Honesty, Punctuality, and Discipline. School A awards ₹x, ₹y, and ₹z to 3, 2, and 1 students respectively, distributing a total of ₹1000. School B awards 4, 1, and 3 students respectively for the same values, distributing ₹1500. School C awards 2, 3, and 1 students respectively, distributing ₹1200. *(i)* Formulate this scenario into a matrix equation $AX = B$. *(1 Mark) (ii)* Find the determinant $|A|$ of the coefficient matrix. *(1 Mark) (iii)* Using the matrix inverse method, find the exact award amount for Honesty, Punctuality, and Discipline. *(2 Marks)* * **Chapter:** Matrices and Determinants * **Source:** [Practice] [Confidence: High]

Q37. Case Study 2: Application of Derivatives (Revenue Optimization) A telephone company in a town has 500 subscribers on its list and currently collects fixed charges of ₹300 per subscriber per year. The company proposes to increase the annual subscription to generate more profit. Market research shows that for every increase of ₹1 in the annual subscription, exactly one subscriber will discontinue the service. Let x be the increase in the annual subscription fee in rupees. *(i)* Write the total revenue function $R(x)$ of the company strictly in terms of x. *(1 Mark) (ii)* Find $R'(x)$ and equate it to zero to find the critical point. *(1 Mark) (iii)* Use the second derivative test to find the increase in subscription x that yields the maximum revenue, and calculate this maximum revenue. *(2 Marks)* * **Chapter:** Application of Derivatives * **Source:** [PYQ 2024] [Confidence: Very High]

Q38. Case Study 3: 3D Geometry (Air Traffic Control) Two flights are flying in the air along two straight-line paths. To avoid collisions, Air Traffic Control (ATC) needs to ensure they maintain a safe minimum distance. Flight 1 follows the path $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$ and Flight 2 follows the path $\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} + (2s+1)\hat{k}$. *(i)* Write the direction ratios of the paths of Flight 1 and Flight 2. *(1 Mark) (ii)* Determine mathematically whether the paths of the two flights are parallel or skew lines. *(1 Mark) (iii)* Calculate the exact shortest distance between the two flight paths to ensure they don't collide. *(2 Marks)* * **Chapter:** Three-Dimensional Geometry * **Source:** [Practice] [Confidence: Very High]

Answer Key

Q1. Principal value branch of $\cos^{-1} x$ is $[0, \pi]$. $\cos(7\pi/6) = \cos(2\pi - 5\pi/6) = \cos(5\pi/6)$. Hence, $\cos^{-1}(\cos(5\pi/6)) = \frac{5\pi}{6}$. **Q2.** $\sin^{-1}(\sin\frac{2\pi}{3}) = \sin^{-1}(\sin(\pi - \frac{\pi}{3})) = \sin^{-1}(\sin\frac{\pi}{3}) = \frac{\pi}{3}$. **Q3.** $|A| = 0$ (Determinant of a skew-symmetric matrix of odd order is always 0). **Q4.** $|adjA| = |A|^{n-1} = 5^{3-1} = 5^2 = 25$. **Q5.** $\frac{dy}{dx} = 2^{\sin x} \cdot \cos x \cdot \log_e 2$. **Q6.** $A = \pi r^2 \implies \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(6)(3) = 36\pi\ \mathrm{cm}^2/\mathrm{s}$. **Q7.** 0 (since $\sin^7 x$ is an odd function). **Q8.** $e^x \sin x + C$. (Using the form $\int e^x(f(x) + f'(x))dx$). **Q9.** Order = 2, Degree = Not defined (due to the $\cos(dy/dx)$ term). **Q10.** I.F. = $e^{\int(1/x)dx} = e^{\log_e x} = x$. **Q11.** $\frac{(\hat{i}-\hat{j})\cdot(\hat{i}+\hat{j})}{|\hat{i}+\hat{j}|} = \frac{1(1)+(-1)(1)}{\sqrt{2}} = 0$. **Q12.** Direction cosines of the z-axis are $0, 0, 1$. **Q13.** $|3A| = 3^3|A| = 27|A|$, so $k = 27$. **Q14.** $\int(3x^2 + 4x^3)dx = x^3 + x^4 + C$. **Q15.** $y' = e^{-x}(2x - x^2)$. Strictly increasing when $x(2-x) > 0 \implies x \in (0, 2)$. **Q16.** $\hat{i}\cdot\hat{i} + \hat{j}\cdot(-\hat{j}) + \hat{k}\cdot\hat{k} = 1 - 1 + 1 = 1$. **Q17.** No. $f(1) = 1$ and $f(-1) = 1$. Two inputs map to the same output. **Q18.** Magnitude = $\sqrt{4+1+4} = 3$. DCs are $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$. **Q19.** Option C (Assertion is true, but Reason is false; continuous functions are not always differentiable). **Q20.** Option A (Both A and R are true, and R is the correct explanation of A; Dot product is $6 - 2 - 4 = 0$).

Q21. Let $\sin^{-1}(3/5) = A \implies \tan A = 3/4$. Let $\cot^{-1}(3/2) = B \implies \tan B = 2/3$. $\tan(A+B) = \frac{3/4+2/3}{1-(3/4)(2/3)} = \frac{17/12}{6/12} = \frac{17}{6}$. **Q22.** $\frac{1}{2}[2(0) - (-6)(5-k) + 1(20-4k)] = \pm 35 \implies 50 - 10k = \pm 70 \implies k = -2, 12$. **Q23.** $\frac{dy}{dx} = e^{-y}(e^x + x^2) \implies e^y dy = (e^x + x^2)dx \implies e^y = e^x + \frac{x^3}{3} + C$. **Q24.** $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta \implies 6 = 12\sin\theta \implies \sin\theta = 1/2 \implies \theta = \frac{\pi}{6}$ (or 30°). **Q25.** In the xy-plane, $z = 0$. So $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{1}{4} \implies x = \frac{3}{2}, y = \frac{5}{4}$. Coordinates: $(\frac{3}{2}, \frac{5}{4}, 0)$.

Q26. Reflexive: $a - a = 0$, divisible by 2. Symmetric: if $a - b = 2k$, $b - a = 2(-k)$. Transitive: $(a-b)+(b-c) = 2k_1 + 2k_2 = 2(k_1+k_2)$, so $a - c$ is divisible by 2. **Q27.** $A = P + Q$ where $P = \frac{1}{2}(A + A^T) = \begin{pmatrix} 3 & 3 \\ 3 & -1 \end{pmatrix}$ and $Q = \frac{1}{2}(A - A^T) = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$. **Q28.** Let $u = x^x \implies u' = x^x(1 + \log x)$. Let $v = (\sin x)^{\log x} \implies \log v = \log x \log(\sin x) \implies v' = (\sin x)^{\log x}\left(\frac{\log(\sin x)}{x} + \log x \cot x\right)$. Add u' and v'. **Q29.** $\int \frac{\frac{1}{4}(4x+6)+\frac{1}{2}}{2x^2+6x+5}dx = \frac{1}{4}\log|2x^2+6x+5| + \frac{1}{2}\int \frac{dx}{2(x^2+3x+5/2)}$. Completing the square gives $\frac{1}{4}\log|2x^2+6x+5| + \frac{1}{2}\tan^{-1}(2x+3) + C$. **Q30.** $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{1}{(1+x^2)^2}$. I.F. = $1 + x^2$. Sol: $y(1+x^2) = \int \frac{1}{1+x^2}dx \implies y(1+x^2) = \tan^{-1}x + C$. **Q31.** $\vec{a}_2 - \vec{a}_1 = \hat{i} + 2\hat{j} + 2\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = -\hat{i} + 2\hat{j} - \hat{k}$. $SD = \frac{|-1+4-2|}{\sqrt{1+4+1}} = \frac{1}{\sqrt{6}}$ units.

Q32. $|A| = 40$. $A^{-1} = \frac{1}{40}\begin{pmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{pmatrix}$. Solving $X = A^{-1}B$ gives $x = 1, y = 2, z = -1$.

Q33. Let radius be r, height h, slant height l (constant). $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(l^2 - h^2)h$. $V'(h) = \frac{1}{3}\pi(l^2 - 3h^2) = 0 \implies l^2 = 3h^2 \implies r^2 = 2h^2$. $\tan\theta = \frac{r}{h} = \sqrt{2} \implies \theta = \tan^{-1}\sqrt{2}$. $V''(h) < 0$ guarantees maximum. **Q34.** Standard application of definite integral limits property. Answer: $-\frac{\pi}{2}\log 2$. **Q35.** Point of intersection is $(4, 4)$. Area = $\int_0^4 (2\sqrt{x} - \frac{x^2}{4})dx = [2 \cdot \frac{2}{3}x^{3/2} - \frac{x^3}{12}]_0^4 = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$ sq. units.

Q36. *(i)* $\begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1000 \\ 1500 \\ 1200 \end{pmatrix}$. *(ii)* $|A| = 3(-8) - 2(-2) + 1(10) = -10 \neq 0$. *(iii)* Solving $X = A^{-1}B$ yields $x = 200, y = 100, z = 200$. **Q37.** *(i)* $R(x) = (500 - x)(300 + x)$. *(ii)* $R'(x) = 200 - 2x = 0 \implies x = 100$. *(iii)* $R''(x) = -2 < 0$. Max revenue $R(100) = 400 \times 400 = ₹1,60,000$. **Q38.** *(i)* Path 1 DRs: $-1, 1, -2$. Path 2 DRs: $1, 2, 2$. *(ii)* Direction vectors are not scalar multiples, so they are not parallel. They are skew lines. *(iii)* $SD = \frac{|(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$. $\vec{a}_2 - \vec{a}_1 = \hat{j} - 2\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = 6\hat{i} + 0\hat{j} - 3\hat{k}$. $SD = \frac{|6|}{\sqrt{36+9}} = \frac{6}{\sqrt{45}} = \frac{2}{\sqrt{5}}$ units.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.